

Aryan Anil Shinde

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EDUCATION

University of Southern California

2025 – 2027

M.S. in Analytics, Los Angeles, CA

Vellore Institute of Technology

2021 – 2025

B.Tech in Computer Science & Engineering

TECHNICAL SKILLS

AI/Automation Tools ChatGPT, GitHub Copilot, Microsoft Copilot, Vertex AI, Power Automate, Prompt Engineering
Workflow/Docs Process Automation, Structured Research Workflows, Source-to-Target Mapping, Metadata Strategy
Languages/Scripting Python (Pandas, NumPy, Scikit-learn), SQL, Git, Java, json_normalize
Reporting/Bi Tools Tableau, Excel, Power BI, dbt, Snowflake, BigQuery
Data Infrastructure GCP, GCS, Pub/Sub, Dataflow, Bigtable, AWS, Kafka, Flink, Confluent Schema Registry
Certifications AWS Cloud Practitioner, Tableau & SQL (DataCamp), Cybersecurity Certification

WORK EXPERIENCE

Integrated Learning Systems India Pvt. Ltd

Jan 2025 – Apr 2025

Data Engineering Intern - ML & Security Analytics

- Orchestrated a production-grade Vertex AI and Kubeflow Pipelines ingestion framework processing 620K network-traffic records across REST APIs, relational databases, and flat files, reaching 97.0% validation accuracy, F1 of 0.9767, and zero training-serving skew.
- Deployed a GCP inference runtime that replicated upstream StandardScaler states from GCS artifacts at serving time, eliminating numeric drift and cutting model prediction error 23% across 8 months of phishing and malware logs.
- Compressed ML pipeline delivery from 14 to 7 days with LLM-driven schema compilation that auto-generated JSON KFP component specs and automated GCS artifact versioning, saving about 40 hours per retraining cycle.

ATCreations Design School

Mar 2024 – Aug 2024

Data & UX Analytics Intern

- Developed a 4-region Tableau BI platform ingesting 180,000+ inventory records across a 3-year window, surfacing a \$0.41 vs. \$0.62/spent marketing efficiency gap and \$275 to \$311 holiday promotional lift for 3 brand accounts.
- Translated 23 stakeholder requirements into SQL-backed reporting frameworks, resolving 6 join mismatches and standardizing 14 calculated-field dependencies, lifting engagement 16% and reducing dashboard refresh errors 34%.
- Delivered weekly Excel pivot-table summaries with VLOOKUPS and conditional formatting to non-technical leadership across 3 brand accounts, enabling KPI tracking without direct database access.

SmartED Innovations

Aug 2023 – Nov 2023

Junior Data Engineer Intern

- Reverse-engineered a brittle ETL codebase with Python/Pandas json_normalize pipelines that processed 850K+ nested JSON payloads into UTF-8 relational dataframes, eliminating cybersecurity and network schema mismatches.
- Authored 40+ Snowflake SQL analytical queries using window functions, CTEs, and multi-tier aggregation across 4M+ monthly monitoring events, reducing analyst query time from 4.2 hours to 18 minutes.
- Validated 3 production pipelines through data quality audits, resolving 1,200+ schema discrepancies across 650K records, reducing downstream data errors 91%, and ranking top 20 out of 100 trainees.

PROJECTS

Real-Time Feature Store | GCP, Pub/Sub, Dataflow, Bigtable

- Architected a dual-layer real-time/offline feature store processing 8M user interaction events per day through Pub/Sub and Dataflow, serving 6 ML models at 12ms p99 read latency via Bigtable and cutting feature staleness from 6 hours to 4 minutes.
- Instrumented skew detection across 42 feature columns, flagging 94% of incidents before production impact and restoring downstream model AUC by 0.031 after remediation.

Data Cost Observability Platform | BigQuery, dbt, INFORMATION_SCHEMA

- Constructed a BigQuery cost observability platform processing 180GB of INFORMATION_SCHEMA metadata monthly, attributing compute spend across 8 analyst teams and surfacing 340+ full-table scans in 30 days.
- Engineered automated query optimization recommendations with dbt macros and slot telemetry, reducing average analyst query runtime from 4.8 minutes to 52 seconds across 1,200 weekly queries.

Streaming Data Contract Enforcement | Kafka, Confluent Schema Registry, Flink, Dataflow

- Designed a real-time data contract layer processing 400K Kafka events/hour, validating Avro payloads with Confluent Schema Registry and Flink within 11ms and routing invalid records to a lineage-rich DLQ.
- Deployed a GCP Dataflow remediation pipeline that reprocessed 2.4M rejected events after schema fixes, recovering 9 days of training data and restoring fraud detection F1 from 0.81 to 0.94 without a full retrain.

LEADERSHIP & INVOLVEMENT

Head Coordinator – BGMI Showdown and MELTDOWN; led end-to-end event planning and execution.

Coordinator – TECH LITE and DESIGNATHON; supported national technical and design events.